A brief review on Unmanned Combat Aerial Vehicle (U.C.A.V)

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Keywords: Unmanned,Aerial,Vehicle,Combat,Drone,UCAV.

1. *Abstract*

An unmanned combat aerial vehicle (U.C.A.V), is an unmanned aerial vehicle (U.A.V) that usually carries aircraft ordinances such as missiles, ATGMs, and/or bombs and is used for drone strikes and other military operations. These UCAVs are always under real-time human control, with varying levels of autonomy. Aircraft of this type have no onboard human pilot keeping the safety point of view. As the operator runs the aircraft from a remote terminal, equipment essential for a human pilot is not needed, resulting in a lower weight and a smaller size compared to a manned aircraft. India, Israel, Turkey, and the United States are widely recognized as industry leaders in U.C.A.V technology. Several other countries have operational domestic U.C.A.Vs and many more have imported armed drones or have development programs underway. In this paper, a comprehensive review of the Unmanned Combat Aerial Vehicle, its components, capabilities, missiles it carries along with its use and future scope has been done.

2. Introduction

UCAV is an aircraft that expands tactical mission options for revolutionary new airpower as an integrated part of a system of systems solution. A UCAV can conduct a mission from ordinary airfields as part of an integrated force package complementary to manned tactical and support assets. The controllers of the UCAV can observe rules of engagement and make critical decisions to use or refrain from using force. The role of the UCAV is "the first day of the war" force enabler which complements a strike package. In this role, UCAVs accomplish preemptive destruction of sophisticated enemy integrated air defenses (IADs) in advance of the strike package and enable the attacking forces by providing reactive suppression against the remaining IADs.UCAVs can provide continuous vigilance with an immediate lethal strike capability to prosecute high value and time-critical targets. A UCAV system can provide "no-win" tactical deterrence against which an enemy's defenses would be ineffective, thereby ensuring air superiority and kill high-value targets with point precession.

3. Comparison of U.C.A.V

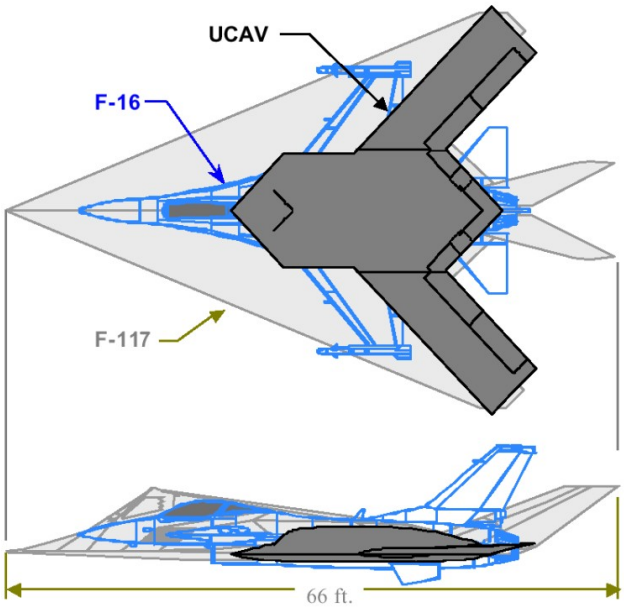


Fig. 1 Comparison of the size of a UCAV with other aircrafts

*F-16 -General Dynamics F-16 Fighting Falcon

*F-117- Lockheed Martin F-117 Nighthawk

*UCAV- Lockheed Martin RQ-170 Sentinel

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Fig. 2 Maiden flight of the XQ-58A Valkyrie

A U.C.A.V may seem like a futuristic fighter aircraft but in reality, it is a lot different. The first and most obvious difference is that it is unmanned. In today's world where we value human life so much and can not bear losses in any form, this piece of technology can bring revolution. Since we need not carry the load of a human, we may use the difference of weight, to carry the equipment or improve on the engine which can prove as a gamechanger in a situation. It can be launched for any action in a very short amount of time. It can do all the activities that any normal fighter aircraft may have performed, but by giving the luxury to the pilot of sitting in a room and controlling the aircraft to do the task assigned. If we compare the size of the UCAV to any other fighter plane, it varies but the models being used in the present are smaller and more compact. They have a very high success rate which tends to grow exponentially if used on a much larger scale.[4]. UAV/UCAV drones are used where there are small areas, areas in remote locations where it is hard to fly. Higher-resolution data can be obtained as compared to traditional aircraft surveying. These drones also provide great efficiency and cost savings for the organizations it is being utilized by while confronting traditional workflows and expectations.UCAVs demand only very small stretches to take off and land, no longer than a few square feet in the case of vertical take-off and landing (VTOL) designs. [9]

4. Components of a UCAV



Fig. 3 Various devices and sensors mounted on a UAV

The components of a UCAV are the same as that of a UAV but with the capability to carry ordinance on board[2][3].

UCAVs have several main components such as:

4.1 Body

The body of a UCAVs comprises a fuselage, plane wings, tail rotor and canopy, multi-rotor frame and arms.

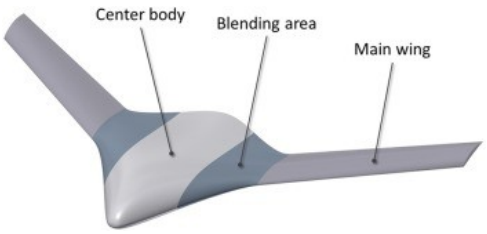


Fig. 4 Body of a UAV

4.2 Power supply with the corresponding platform

Smaller UCAVs usually fly on batteries while the larger ones use fuel or even solar power(Not being used very widely as of now).Most of the UCAVs used for traditional purpose use engines such as or comparable to Pratt & Whitney F100-PW-220U(Fig.5) engine used by X-47B Pegasus.



Fig. 5 Pratt & Whitney F100-PW-220U

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4.3 Computer operations

UCAVs' hardware systems are constantly being specialized, operation numbers increased and accelerated to support the operating system with no failures.

4.4 Sensors

It mainly has three types of sensors:

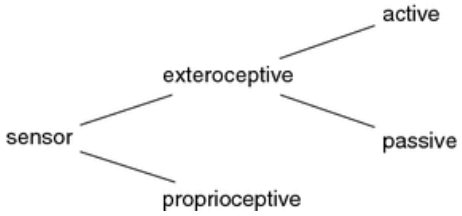


Fig. 6 Classification of sensors

> *Proprioceptive*:

Measurements of movement relative to the robot's internal frame of reference (also called dead reckoning).

> *Exteroceptive*:

Measurements of the layout of the environment and objects relative to a robot's frame of reference.

> *Exproprioceptive*:

Measurement of the position of the robot body or parts relative to the layout of the environment[31].

4.5 Actuators

Installed actuators are determined by the type of UCAVs (several electronic controllers, engines, propellers and others).



Fig. 7 Northwest UAV Pegasus Actuator

4.6 Software

The uninterrupted flight is enabled with installed software safely leading the UCAVs on its way and giving the information where to go and when to react.

4.7 Loop principles

It has open loops for elementary types and closed loops for larger and more sophisticated systems.

4.8 Flight controls

It has similar flight controls like aerial vehicles flown by a pilot. However, automatic flight control is much more demanding.

4.9 Telecommunications

The connection is quite regular (antenna and analog-digital converter) and enables the transmission of data needed for the flight to proceed. As the radio signal can be transmitted from ground control, a remote system as well as another manned aerial vehicle.

4.10 UCAV software

UCAV software is designed to tell the UCAVs where to go and what to do while flying from A to B. To understand and connect all necessary information of the UCAVs, the software part becomes a very complex system. The software installed in the UCAVs operates in a layer like a system. Furthermore, the layers are divided into tiers that perform in various time slots. The layers have to be combined properly to control the flight patterns, altitude and other important information for the UCAVs to work and act accurately. These combinations of layers are called the flight stack or autopilot. The information received has to be analyzed during the flight. To achieve a unified component's communication, a generic architecture must be designed and promoted.

4.11 Flight control system

Current flight control is basically in this behavior layer. Some advanced control arithmetic is also realized in this layer. In this paper, as far as system nonlinear strong-coupling characters, corresponding control rules are designed by the use of a dynamic inversion technique [6]. It resolved the control problem when the speed changes as the result of flight at an angle of attack, and sideslip angle, roll angle. The flight control system structure of nonlinear dynamic inversion is shown in Figure 4.

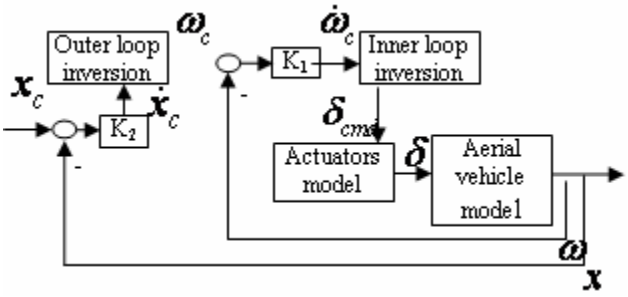


Fig. 8 Flight control system structure

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4.12 Software layers are:

· *A Firmware*: It operates from machine code to processor and afterward to memory access.

· *A Middleware*: It conducts flight control, navigation, and telecommunication.

· *An Operating System*: It monitors optic flow, avoids interference while Simultaneous Localization & Mapping (SLAM) searches for the solution and decides what the action according to received information would be.

Requirements of firmware and middleware are time-critical. The requirement of the operating system is computer-intensive.

4.13 Autonomous control structure

Based on the artificial brain theory [6][15], an autonomous control system may be classified into three layers namely:-

4.13.1 *Perception fusion layer:*

In this layer, the information is transmitted from a multi-sensors system to a relative system and was fused ultimately.

4.13.2 *Cognition and decision layer:*

In this layer, a cognitive model was designed based on three-phase memory mechanisms that had ultra short-time memory and long-time memory.

4.13.3 *Behavior control layer:*

In this layer, a planning function was added that increased behavior intelligence. The flight control was simulated using the dynamic inversion method. And the result showed the design of the execution sub-system was reasonable through tracking the mobile trajectory.

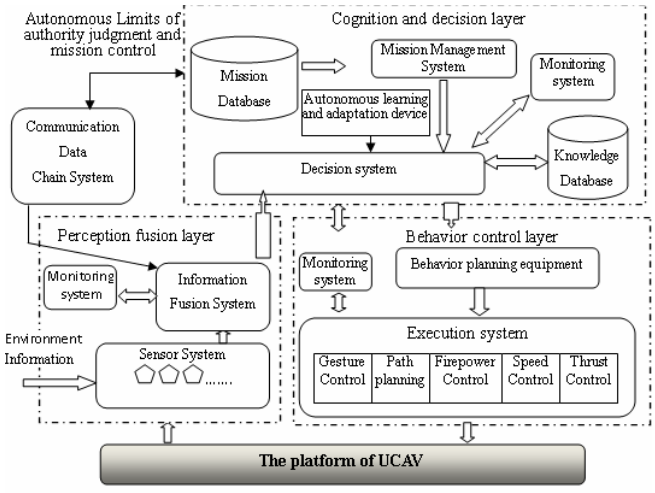


Fig. 9 UCAV autonomous control structure

An autonomous control structure based on cognitive science is designed, as showed in Fig. 9 and it is divided into three layers which are perception fusion layer, cognition and decision layer, and behavior control layer. UCAV could perceive online situations, execute assigned missions autonomously and make a decision during aviation in principle.[28][6]

4.14 Weapons on Board:

There are several types of weapons that are carried by different UCAVs:-

Here examples from the General Atomics MQ-9 Reaper Drone have been taken, which has proved its operational capability in many of the operations.[7]

4.14.1 *AGM-114 Hellfire:-*

The AGM-114 Hellfire (Fig. 10) is an air-to-surface missile (ASM) first developed for anti-armor use, but later models were developed for precision drone strikes against other target types, and have been used in a number of targeted killings of high-profile individuals.



Fig. 10 AGM-114 Hellfire Missile

4.14.2 *GBU-12 Paveway II:-*

The GBU-12 Paveway II (Fig. 11) is an American aerial laser-guided bomb, based on the Mk 82 500-pound general-purpose bomb, but with the addition of a nose-mounted laser seeker and fins for guidance. A member of the Paveway series of weapons, Paveway II entered into service c. 1976.



Fig. 11 GBU-12 Paveway Missile

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4.14.3 *MBDA Brimstone:-*

MBDA Brimstone (Fig. 8) is an air-launched ground attack missile developed by MBDA.



Fig. 12 A Brimstone missile

5. Parameter for classifcation of a UCAV

5.1 Types:-

A UCAV can be classified into two types:-

5.1.1 MALE or Medium Altitude Long Endurance UCAV:

It can fly up to an altitude of 30,000 feet and travel over 200 km.

5.1.2 HALE or High Altitude Long Endurance UCAV:

It can go beyond 30,000 feet and has a range of a few thousand.

5.2 *Speed :-*

Mach* 0.18 (222 kmph)-Mach 1 (1,234 kmph)
*Ratio of a drone's speed to speed of sound.
*This figure is never specific and changes from model to model.

5.3 *Cost of manufacturing:-*

Here is the apprixiamate cost of one of the most videly used drone, the General Atomics MQ-1 Predator UCAV.

Program cost: US $2.38 billion (2011)

Unit cost: US $4.03 million (2010)

* This figure is approximate and varies as per the aircraft model.

5.4 *Payload carrying capacity:-*

It depends from aircraft to aircraft.

The X-47B has a payload capacity of 2,000kg, a wingspan of 18.9m, maximum gross take-off weight of 19.95t. It is powered by a Pratt & Whitney F100-PW-220U engine.



Fig .13 A X-47B Pegasus refueling mid air

7. Uses of U.C.A.V's

7.1 *Fighting Terrorism*

UCAVs have played a vital role in fighting terrorism in terror-stricken regions.A MQ-1 Predator UAVs armed with Hellfire missiles (Fig.14) have been used by the U.S. as platforms for hitting ground targets. Between the years 2006 and 2009, UCAV-launched missiles allegedly had killed between 750 and 1,000 people in Pakistan occupied regions, according to the report. Out of these, about 20 people were said to be leaders of various terror groups. As per the report Overall, 66% to 68% of the people killed were militants, and 31% to 33% were civilians. U.S. officials disputed the percentage for civilians.[1]

7.2. *Killing a High-Value Target in Foreign Territory*

UCAVs can and are being used to take down high-value targets with pinpoint precision in foreign territory. They make less noise which makes them less vulnerable to being spotted and leave the region quickly. A top Irani General, Qasem Soleimani was struck on 3rd January 2020 by U.S. UCAV strike. An MQ-9 Reaper UCAV (Fig.14) fired at least two Hellfire missiles at vehicles carrying Iranian Major General Qassem Soleimani, Iraqi militia leader Abu Mahdi al-Muhandis and their entourage upon arrival at Baghdad International Airport.[1]



Fig. 14 A MQ-1 Predator in Afghanistan equipped with AGM-114 Hellfire missile

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7.3. *Increasing the operational range of the Navy*

The navy may use it to increase its range and launch it from an aircraft carrier(if a carrier-based version is used). This would also decrease the response time since it needs no pilot to dress up to take charge. It may be operated from the same operations room from where the rest of the aircraft carrier is being operated and engagements are been ordered. Even if the drone is lost, no rapid rescue is needed since it is unmanned.[1][26]

7.4. *Air Superiority and Air Supremacy*

The few moments at the start of any conflict would decide which side gains the advantage throughout. In this matter, the UCAV would prove crucial since it may be used in long-range strikes to destroy the air defense capability so as the paratroopers could be dropped, or in close quarters for air-to-air conflicts so as to respond immediately, preventing the enemy from penetrating deep inside the territory before the principle fighters and bombers could arrive to suppress the enemy.[1]

7.5 *Electronic Warfare*

Any compact electronic warfare equipment may be outfitted along with other ordinances so as to be used to wreak havoc on enemy intelligence by using electronic warfare, in which we deny the opponent the advantage of and ensure friendly unimpeded access to, the EM spectrum. Due to this, the enemy may not be able to use its radar and other critical equipment, to coordinate assaults or other operations planned prior. This would bring down the command structure and would make the gathering of intelligence difficult for the enemy and provide an advantage to the party making use of this technology.[1]



Fig. 15 MQ-4C-Triton

7.6 *Healthcare in Hostile Regions*

Since the early days, medicine has had a profound impact on preventing disease, increasing life expectancy, and raising general standards of living. However, many hostile regions lack access to even basic healthcare as they do not have a stable healthcare service due to ever-mounting tensions inside the country most of the time. While medical supplies can be delivered by traditional means, certain circumstances call for quick and safe access to drugs, blood, and medical technology for the war-stricken public and soldiers— a need UCAVs could fulfill.[8][29]

8. Future Scope

Nowadays several organisations are working on developing a drone aircraft carrier which could help UCAVs launch and later retrive them upon the completion of the task at any location around the globe.Future work on the autopilot system should have an integrated GPS module for better positioning of the drone. The GPS and IMU can be fused using an estimator based approach, to get somewhat accurate position data. Future versions of the autopilot should also have the ability to transmit light data to a ground-based database system for monitoring, analysis and for future research and development. This should be done by including a 900 MHz radio modem, in the autopilot loop. [23]

9. Conclusion

In this paper, the procedure to built UCAV has been perspicuously stated. The small scale UCAV is cost-effective and has many applications in various fields like military, intelligence,etc. This brief look at the UCAV and other emerging technologies merely scratched the surface of employing UCAVs in a strike role. Further research should focus on control of the CAS level. Additionally, further study must decide the proper manned-unmanned force mix to meet future battlefield requirements, as well as doctrinal and the tactical operational considerations for employing UAVs in civil airspace and in concert with Air Forces.

Compliance with ethical standards

Conflict of interest: The authors declare that they have no conflict of interest.

Funding: No funding was acquired.

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